(continued)

"During the development activity, specific testing was done to ensure that the mathematical equations and logic paths provided the results expected. These algorithms and logic paths were checked for accuracy and, where possible, compared against results from external sources and against the system design specification (SDS)."[18]

At a more technical level, this approach encourages a clean, modular design with well-defined interactions (or APIs, if you prefer a more technical term) between the modules. It also results in code that's reliable, accurate, and extremely well tested.

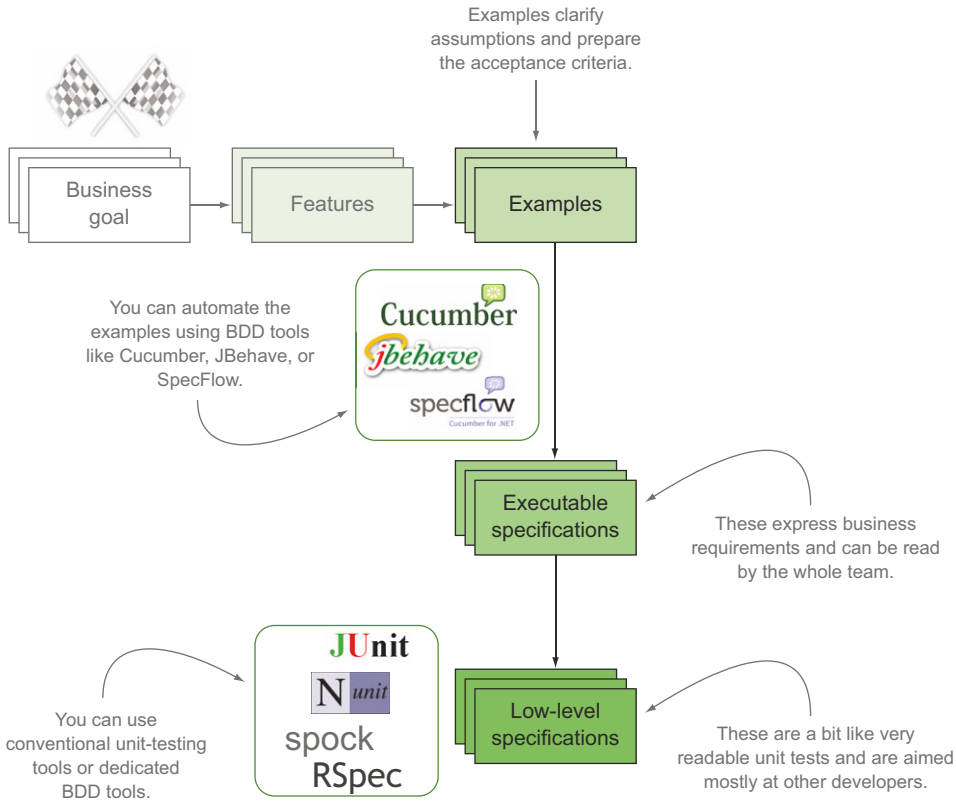


Figure 1.10 High-level and low-level executable specifications are typically implemented using different tool sets.

[18] William A. Madden and Kyle Y. Rone, "Design, Development, Integration: Space Shuttle Primary Flight Software System," *Communications of the ACM* (September 1984).